

Lesson 5 - Quick Reference Card

Database Setup

```
# .env file
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=laravel_db
DB_USERNAME=root
DB_PASSWORD=
```

Migration Commands

```
# Create migration
php artisan make:migration create_products_table

# Create Model with Migration
php artisan make:model Product -m

# Run migrations
php artisan migrate

# Rollback
php artisan migrate:rollback

# Refresh all
php artisan migrate:fresh

# With Seeding
php artisan migrate:fresh --seed

# Show status
php artisan migrate:status
```

■ Creating Table

```
Schema::create('products', function (Blueprint $table) {
    $table->id();
    $table->string('name');
    $table->decimal('price', 10, 2);
    $table->integer('stock')->default(0);
    $table->timestamps();
});
```

■ Common Column Types

```
// Numbers
$table->id(); // PRIMARY KEY
$table->integer('votes');
$table->decimal('price', 8, 2);
$table->unsignedBigInteger('user_id');

// Strings
$table->string('name'); // VARCHAR(255)
$table->string('email', 100); // VARCHAR(100)
$table->text('description');
$table->json('options');

// Date & Time
$table->date('birth_date');
$table->datetime('created_at');
$table->timestamp('verified_at');
$table->timestamps(); // created_at + updated_at

// Special
$table->boolean('is_active');
$table->enum('status', ['active', 'inactive']);
```

■ Modifiers

```
$table->string('email')->nullable();
```

```
$table->string('name')->default('Guest');
$table->integer('votes')->unsigned();
$table->string('email')->unique();
$table->string('phone')->after('email');
```

Foreign Keys

```
// Modern way
$table->foreignId('user_id')
    ->constrained()
    ->onDelete('cascade');

// Traditional way
$table->unsignedBigInteger('user_id');
$table->foreign('user_id')
    ->references('id')
    ->on('users')
    ->onDelete('cascade');

// onDelete options
->onDelete('cascade')    // Delete
->onDelete('set null')   // NULL
->onDelete('restrict')   // Prevent
```

Modifying Tables

```
# Create modification migration
php artisan make:migration add_phone_to_users_table
```

```
// Add column
Schema::table('users', function (Blueprint $table) {
    $table->string('phone')->nullable();
});

// Modify column (requires doctrine/dbal)
composer require doctrine/dbal
```

```
Schema::table('users', function (Blueprint $table) {
    $table->string('name', 100)->change();
});

// Drop column
Schema::table('users', function (Blueprint $table) {
    $table->dropColumn('phone');
});

// Rename
Schema::table('users', function (Blueprint $table) {
    $table->renameColumn('old_name', 'new_name');
});
```

□ Database Seeding

```
# Create seeder
php artisan make:seeder UserSeeder

# Run seeders
php artisan db:seed
php artisan db:seed --class=UserSeeder
```

```
// In Seeder
public function run(): void
{
    User::create([
        'name' => 'Admin',
        'email' => 'admin@example.com',
        'password' => Hash::make('password'),
    ]);

    // Or using Factory
    User::factory()->count(50)->create();
}
```

❑ Model Factories

```
php artisan make:factory ProductFactory
```

```
public function definition(): array
{
    return [
        'name' => fake()->words(3, true),
        'price' => fake()->randomFloat(2, 10, 1000),
        'stock' => fake()->numberBetween(0, 100),
    ];
}

// Usage
Product::factory()->create();
Product::factory()->count(50)->create();
```

❑ Indexes

```
$table->string('email')->unique();
$table->index('email');
$table->index(['first_name', 'last_name']);
$table->fullText('description');

// Drop
$table->dropIndex('users_email_index');
$table->dropUnique('users_email_unique');
```

❑ Checking

```
if (Schema::hasTable('users')) {
    // exists
}

if (Schema::hasColumn('users', 'email')) {
    // exists
}
```

```
}
```

□ Best Practices

□ Use clear table names □ nullable only when needed □ Use Foreign Keys □ Add Indexes for search □ Always write down()

□ Don't modify old migrations □ Don't use migrate:fresh in production

□ Quick Links

- [Main Lesson](#)
- [Next Lesson](#)